

HACKHUB – Problem Statements

SENIOR CATEGORY:(For 3rd and 4th years)

Problem Statement 1: Tracking Packages from Arrival to Departure in a Cost-Effective Way.

Package tracking is the process of monitoring the movement of shipments from the moment they arrive at a facility until they reach their final destination. It ensures smooth logistics operations, minimizes delays, and optimizes costs. A well-implemented tracking system provides **real-time updates**, allowing businesses and customers to know the exact location and status of their packages at any given time. It involves **data management**, where key details like package size, weight, destination, and expected delivery time are recorded. Efficient tracking ensures **cost optimization** by selecting the best routes and resources, reducing unnecessary expenses. Additionally, it helps in **error reduction**, preventing misplacements or delays through proper scanning and verification at multiple checkpoints. By integrating **automated alerts and notifications**, businesses can enhance efficiency and improve customer satisfaction. A cost-effective package tracking system not only speeds up delivery operations but also ensures reliability, making logistics more streamlined and productive.

Problem Statement 2: Virtual Avatar for Passenger Engagement and Airport Assistance.

A virtual avatar is an AI-powered assistant designed to enhance passenger experience at airports. It provides real-time information about airport facilities, services, and navigation. The avatar can guide passengers to gates, baggage claim areas, and security checkpoints, reducing confusion. It offers details on restaurants, shopping areas, medical services, and Wi-Fi availability. Using natural language processing (NLP), it understands and responds to passenger queries efficiently. The avatar also provides flight updates, boarding information, and weather conditions to keep travellers informed. It can engage in general conversations, answering FAQs and offering travel tips. The system can be integrated into mobile apps, or touchscreens for easy access. Advanced features like multilingual support improve accessibility for international travellers. AI-driven recommendations enhance personalization based on passenger needs. The avatar reduces wait times at information desks, improving efficiency. Facial recognition can be integrated for personalized assistance. It operates 24/7, ensuring continuous support without requiring human staff. This technology enhances airport operations and ensures a seamless travel experience. By implementing a virtual avatar, airports can improve passenger satisfaction and operational efficiency.

Problem Statement 3: Count of cars entering/exiting, Record ANPR, record video snippet for review, track duration of parking, loitering violations, identify incidents among others.

A Landside/Airport Parking Monitoring System is designed to efficiently track and manage vehicle movement within airport parking areas. It ensures smooth operations by monitoring the entry and exit of vehicles, maintaining records, and enhancing security. The system records the count of cars entering and exiting in real time, helping in traffic flow management. It integrates Automatic Number Plate Recognition (ANPR) technology to identify and log vehicle details for security and auditing purposes. Additionally, video snippets are recorded for review, ensuring a reliable record of activities within the parking area. The system also tracks the duration of parking, allowing for better enforcement of time limits and reducing congestion. It can detect loitering violations, identifying vehicles that stay beyond allowed durations without valid reasons. Incident detection features help recognize unauthorized activities, accidents, or security breaches, allowing prompt responses.

By automating parking management, this system enhances **efficiency, security, and compliance** while reducing manual intervention. It ensures smooth airport parking operations by preventing unauthorized parking, improving safety, and optimizing space utilization.

Problem Statement 4: Need a collaborative system to share information, receive approval and real-time updates from factory to end.

The cargo and logistics industry relies on smooth communication and real-time updates to ensure efficient operations. However, the lack of a centralized system often leads to delays, miscommunication, and inefficiencies. Important information is scattered, approvals take too long, and shipment status updates are not always clear. To solve this, there is a need for a **collaborative system** where manufacturers, logistics providers, and end-users can **easily share information, get quick approvals, and receive real-time updates** from the factory to the final delivery. A digital platform with **live tracking, automated notifications, and approval workflows** will improve transparency, speed up processes, reduce delays, and ensure a smoother supply chain experience for all stakeholders.

Problem Statement 5: Identify and classify traffic type - buses, cars, SUVs per day.

Managing urban traffic efficiently requires an accurate understanding of the types and volume of vehicles on the road. However, without an automated system, tracking and classifying different vehicle types, such as buses, cars, and SUVs, becomes challenging. The main objective is to develop an intelligent traffic monitoring system that uses video analytics to automatically identify, classify, and count vehicles in real time. This data will help city planners and traffic authorities improve traffic flow, optimize infrastructure, and enhance road safety through better decision-making.

Problem Statement 6: Retail video analytics to increase penetration to duty free.

Duty-free stores often struggle to attract and engage more customers, missing key insights into shopper behaviour. Without proper analytics, it is difficult to understand foot traffic patterns, customer preferences, and sales conversion rates. The main objective is to use video analytics to track customer movement, analyse shopping trends, and optimize store layouts to increase footfall and sales in duty-free zones. This technology will help retailers improve customer engagement, refine marketing strategies, and create a more effective shopping experience, ultimately boosting sales and revenue.

Problem Statement 7: Medicine Reminder & Tracker

Develop an app that reminds patients to take their medication on time and track their adherence. Medication non-adherence is a significant global health issue, leading to poor treatment outcomes, increased hospitalizations, and higher healthcare costs. Many patients, especially those managing chronic illnesses, struggle to take their medications on time due to forgetfulness, complex schedules, or lack of awareness. A mobile/web app that provides personalized medication reminders, medicine identification, adherence tracking, and real-time alerts can help patients stay on track with their prescribed treatments. By integrating features such as AI-driven health insights, progress reports, and integration with wearables or electronic health records, this solution empowers patients to manage their health more effectively while enabling healthcare providers to monitor adherence and intervene when necessary.

Problem Statement 8: Virtual Science Lab

Create a simple simulation-based app or website to conduct basic science experiments. This can help students visualize concepts better, especially if they don't have access to a well-equipped lab. Many students, particularly those in underprivileged or remote areas, lack access to well-equipped science laboratories, limiting their ability to conduct hands-on experiments. This gap in practical learning can make it difficult to fully grasp scientific concepts and hinder engagement in STEM education. A simulation-based app or website can bridge this gap by providing an interactive digital platform where students can conduct virtual experiments in physics, chemistry, and biology. By offering realistic simulations, gamified learning experiences, and accessibility across multiple devices, this solution aims to enhance conceptual understanding, foster curiosity, and make science education more engaging and inclusive.

Problem Statement 9: A speech-to-text solution is required for automatic call logging of complaints from the terminal. This will also help in creating a dashboard at APOC.

A speech-to-text solution is required to automatically log complaints received via calls from the terminal. This system will convert spoken complaints into text, eliminating the need for manual entry. The transcribed complaints will be stored in a database and categorized based on issue type, urgency, and location. Additionally, this solution will help in creating a dashboard at APOC (Airport Operations Control). The dashboard will display real-time complaint data, including the number of complaints received, their current status, and trends over time. This will enable the APOC team to monitor and address issues efficiently, improving response time and overall airport operations.

Problem Statement 10: AI-BASED ONLINE EXAM CHEATING DETECTION

Online exams are vulnerable to various forms of cheating, such as screen sharing, unauthorized device usage, and impersonation. An **AI-powered proctoring system** can help ensure exam integrity by using computer vision and machine learning to monitor candidates in real time. This system can detect **suspicious eye movements, multiple faces in the frame, and the presence of unauthorized devices**. By analyzing behavioral patterns and flagging anomalies, it helps prevent cheating and ensures a fair examination environment.

JUNIOR CATEGORY:(For 1st and 2nd years)

Problem Statement 1: Create a web application that educates people on how to recycle different materials effectively.

A Recycling Awareness Platform is a digital app that educates users on proper recycling methods. It provides detailed guidelines on recyclable and non-recyclable materials, helping users reduce waste contamination. The app offers step-by-step instructions for recycling plastic, paper, glass, metal, and electronics. Users can scan items using AI-powered recognition to get instant disposal guidance.

The application promotes environmental consciousness and sustainable waste management. It integrates reminder notifications for scheduled waste disposal. Educational videos and tutorials make learning fun and effective. By encouraging responsible recycling, the platform helps reduce landfill waste. Eco-friendly challenges and leaderboards motivate users to stay active. The app supports a greener future through smarter waste disposal.

Problem Statement 2: Smart Assessment Board

A Smart Assessment Board is a digital solution designed to help teachers conduct quiz competitions easily and efficiently. In this case, a school teacher wants to organize a quiz on identifying states and their capitals. The system provides an interactive and automated platform to manage the competition smoothly. The board can feature multiple-choice questions, buzzer-based answering, and instant result evaluation. It may include a touchscreen interface or a web-based platform for students to participate seamlessly. Automated scoring and real-time leaderboard updates ensure fair and engaging competition. The system supports customizable quizzes, allowing teachers to modify content based on different subjects. It enhances student engagement, reduces manual effort, and ensures a fun learning experience. A Smart Assessment Board makes quizzes interactive, efficient, and enjoyable for both teachers and students.

Problem Statement 3: Movie Night Planner

Develop a Movie Night Planner web app that simplifies the process by recommending movies, snack pairings, and discussion topics based on user preferences. The app will allow users to customize their movie night by selecting themes, moods, or genres while also suggesting complementary snacks and conversation starters. This will enhance the overall experience, making movie nights more fun, interactive, and engaging.

Problem Statement 4: Advanced E-Commerce with Augmented Reality

Augmented Reality (AR) in shopping malls enhances the traditional shopping experience by allowing customers to virtually try on clothing and accessories in real-time. This technology provides accurate size and fit simulations, reducing the need for physical trials. Shoppers can explore a diverse range of virtual products, improving convenience and variety. AR seamlessly integrates with e-commerce platforms, enabling users to purchase items directly after virtual trials. It enhances customer engagement by offering interactive and personalized shopping experiences. The system reduces returns and exchanges by ensuring better size accuracy. Shoppers can also receive real-time recommendations based on preferences and previous purchases. This innovation bridges the gap between online and offline shopping, making it more immersive and efficient. Retailers benefit from higher conversion rates and improved customer satisfaction. Augmented Reality in e-commerce redefines the shopping experience, combining technology with convenience.

Problem Statement 5: Develop a smart travel assistant that provides real-time information, personalized recommendations, and language translation to enhance tourist convenience and navigation.

Design an innovative and cost-effective digital solution that provides real-time, reliable information on local attractions, accommodations, and transportation while overcoming language barriers and offering personalized recommendations. The solution should enhance the overall travel experience by enabling seamless navigation, cultural understanding, and efficient trip planning for tourists.

Problem Statement 6: Design an AI-powered web platform that uses facial recognition to automate attendance tracking in classrooms or offices.

Traditional attendance systems are time-consuming and prone to proxy attendance. The system should recognize faces from a live camera feed, mark attendance automatically, and generate reports in real-time.

Problem Statement 7: Develop a web platform where users can participate in fitness challenges, track progress, and compete with others.

Many people struggle to stay motivated in their fitness journey. The platform should allow users to join daily/weekly fitness challenges, log workout progress, and earn points/badges for consistency. Admins can create custom challenges, and users can interact via leader boards and forums.

Problem Statement 8: Develop a web-based system to coordinate relief efforts by connecting NGOs, volunteers, and affected individuals during disasters.

During disasters, relief efforts are often unorganized, leading to inefficient resource distribution. This platform should allow NGOs to list required aid, volunteers to register, and donors to contribute. A real-time dashboard should track relief operations, available resources, and affected areas.

Problem Statement 9: Create an ML-integrated web platform that optimizes last-mile delivery by predicting the best delivery routes and schedules.

Delivery services face inefficiencies due to poor route planning. This system should analyse real-time traffic, delivery locations, and weather conditions to suggest optimized delivery schedules and reduce fuel costs. A reinforcement learning model can improve route efficiency over time.

Problem Statement 10: Build a marketplace that connects farmers directly to buyers, reducing middlemen and ensuring fair prices.

Develop a web-based marketplace that directly connects farmers with buyers (retailers, wholesalers, and consumers). The platform should enable farmers to list their products, set prices, and receive orders without intermediaries.